

RAJALAKSHMI ENGINEERING COLLEGE

LAB MANUAL : CYBER AND DIGITAL FORENSICS CR23532

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING CYBER SECURITY

DONE BY

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EXP 1 **Windows Forensics – Disk Imaging & Analysis**

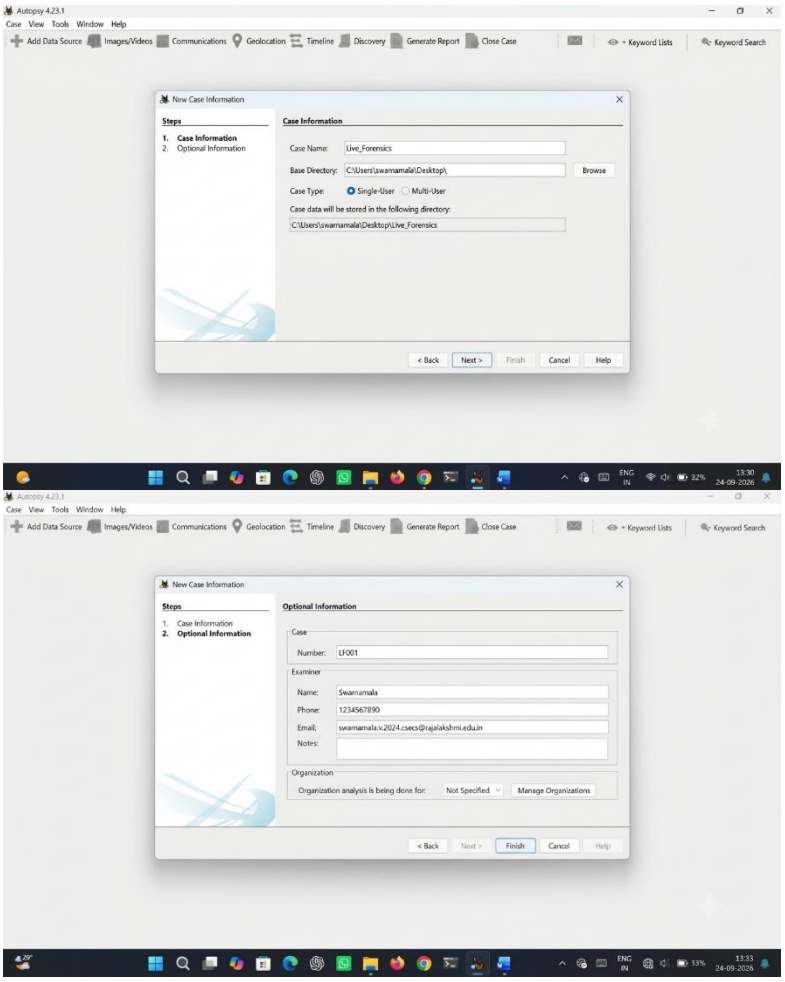
AIM:

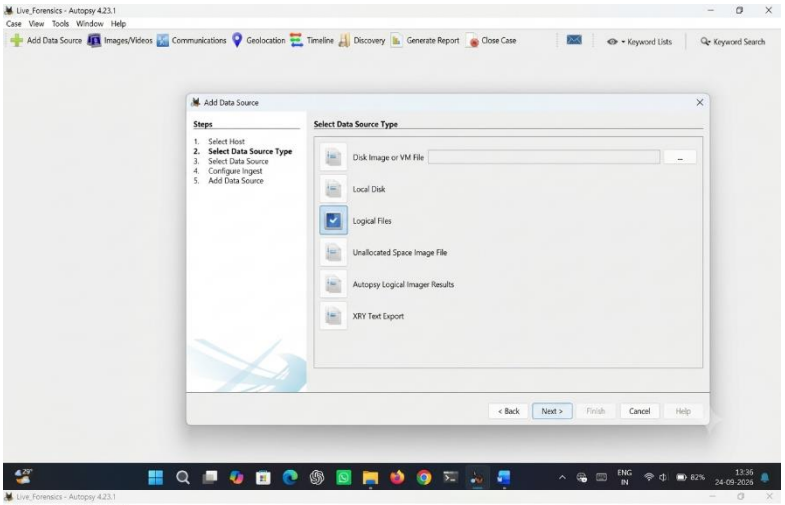
To create a forensic image of a storage device and analyze it using FTK Imager and Autopsy.

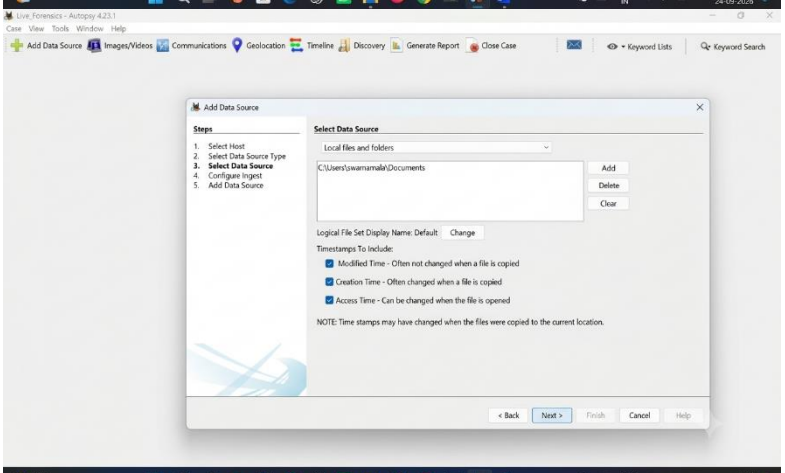
PROCEDURE:

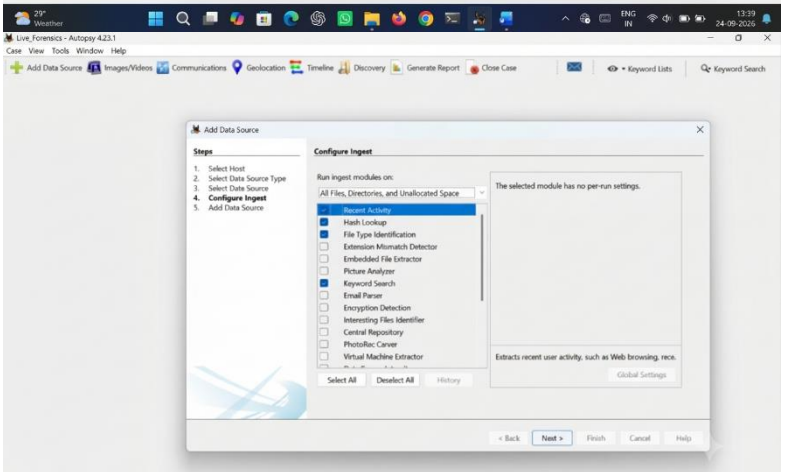
1. Connect the storage device to the system.
2. Open FTK Imager.
3. Select **File → Create Disk Image**.
4. Select the appropriate source type and source drive.
5. Select the destination location for the forensic image.
6. Enter the case details and image filename.
7. Select **E01 (Expert Witness Format)**.
8. Enable verification and hash calculation.
9. Click **Start** to begin the acquisition.
10. Wait until the imaging process is completed and note the generated MD5 and SHA1 hash values.
11. Open the forensic image using Autopsy for analysis.

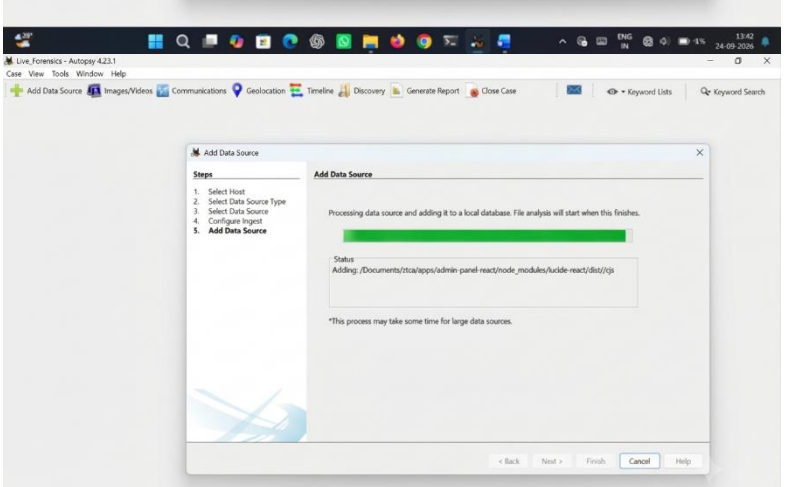
OBSERVATION:

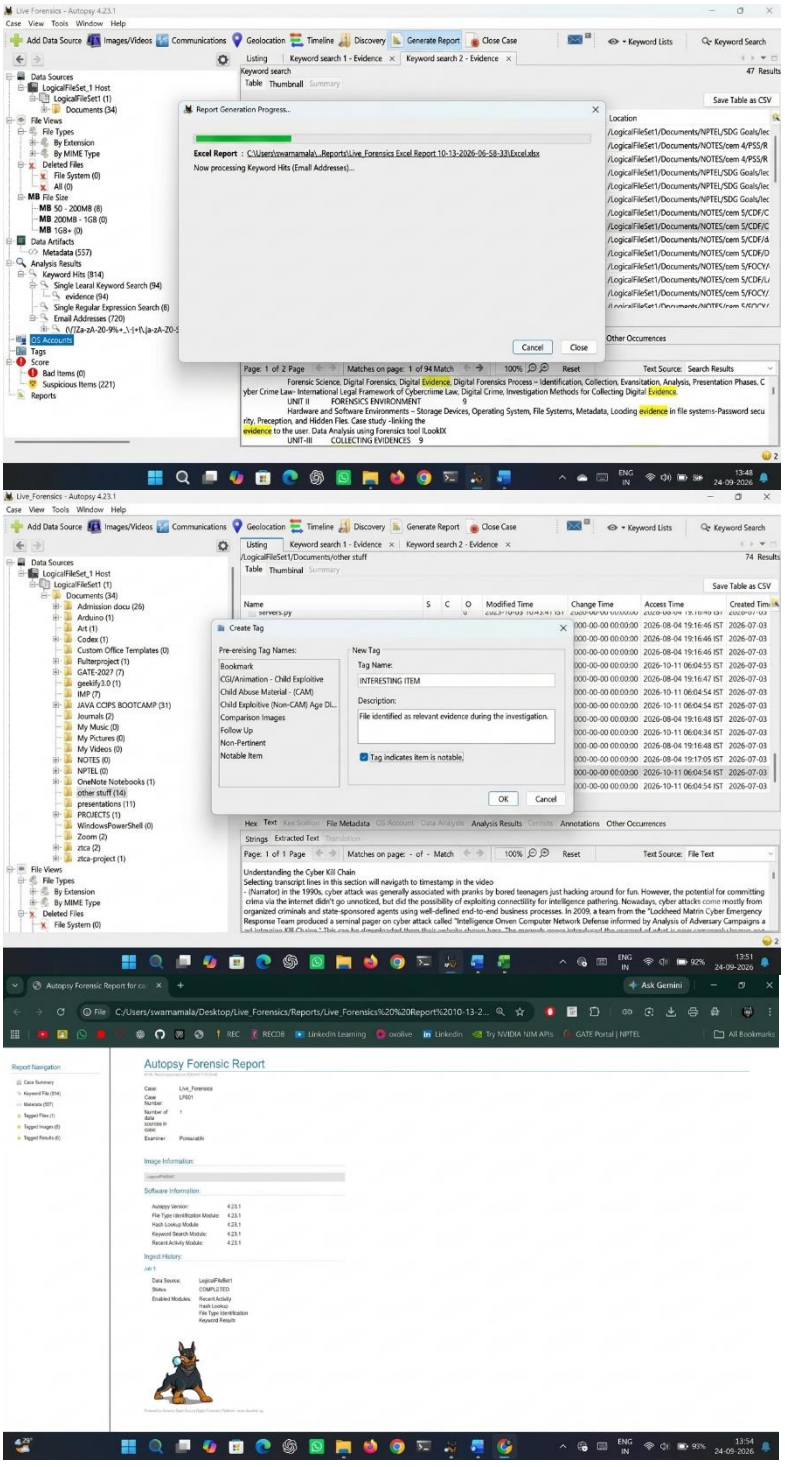



Live_Forensics - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back Next > Finish Cancel Help


Live_Forensics - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source
Local files and folders
C:\Users\swarnamala\Documents
Add
Delete
Clear
Logical File Set Display Name: Default
Change
Timestamps To Include:
Modified Time - Often not changed when a file is copied
Creation Time - Often changed when a file is copied
Access Time - Can be changed when the file is opened
NOTE: Time stamps may have changed when the files were copied to the current location.
< Back Next > Finish Cancel Help


Live_Forensics - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Configure Ingest
Run ingest modules on:
All Files, Directories, and Unallocated Space
Recent Activity
Hash Lookup
File Type Identification
Extension Mismatch Detector
Embedded File Extractor
Picture Analyzer
Keyword Search
Email Parser
Encryption Detection
Interesting Files Identifier
Central Repository
PhotoRec Carver
Virtual Machine Extractor
The selected module has no per-run settings.
Extracts recent user activity, such as Web browsing, rece.
Global Settings
Select All
Deselect All
History
< Back Next > Finish Cancel Help


Live_Forensics - Autopsy 4.23.1
Case View Tools Window Help
Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case
Keyword Lists Keyword Search
Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Add Data Source
Processing data source and adding it to a local database. File analysis will start when this finishes.
Status
Adding: /Documents/rtca/apps/admin-panel-react/node_modules/lucide-react/dist//cjs
*This process may take some time for large data sources.
< Back Next > Finish Cancel Help



The forensic image was created successfully and the generated MD5 and SHA1 hash values were displayed for verification.

RESULT:

Thus, the forensic image was successfully created and verified using FTK Imager.

Exp 2 **FTK Imager – Disk Imaging and Evidence Acquisition**

AIM:

To acquire a forensic image of a storage device using FTK Imager while preserving the integrity of digital evidence.

PROCEDURE:

1. Launch FTK Imager as Administrator.
2. Select File → Create Disk Image.
3. Choose the source type as Physical Drive and click Next.
4. Select the required source drive and click Finish.
5. Click Add to specify the destination.
6. Choose E01 (Expert Witness Format) and click Next.
7. Enter the case details (Case Number, Evidence Number, Examiner, Description).
8. Specify the destination folder and image filename.
9. Enable Verify images after acquisition, Calculate MD5 and Calculate SHA1.
10. Click Start to begin the acquisition.
11. Wait until the imaging completes successfully.
12. Note the generated MD5 and SHA1 hash values and check that the verification status shows SUCCESS.

OBSERVATION:

	Filename	Full Path	Size (bytes)	Created	Modified	Accessed	Is Deleted
2	[root]	NEW VOLUME [FAT	12288				no
3	VBR	NEW VOLUME [FAT	512				no
4	reserved sectors	NEW VOLUME [FAT	18944				no
5	[unallocated space]	NEW VOLUME [FAT	0				no
6	FAT1	NEW VOLUME [FAT	1970688				no
7	FAT2	NEW VOLUME [FAT	1970688				no
8	dhs nov14	NEW VOLUME [FAT	4096	2020-Aug-07 19:34:4	2020-Aug-07 19:34:4		no
9	f9-AC~1.JPG	NEW VOLUME [FAT	28749	2016-Jul-07 20:27:5(	2016-Jul-07 20:27:5(		yes
10	Canara bank - SME -	NEW VOLUME [FAT	4096	2012-Oct-05 11:22:5	2012-Oct-05 11:22:5		no
11	rpt1	NEW VOLUME [FAT	4096	2016-Nov-07 18:28:1	2016-Nov-07 18:28:1		no
12	SREE BHAVANI (27-	NEW VOLUME [FAT	4096	2006-Nov-02 5:01:0(	2006-Nov-02 5:01:1(		no
13	MA0070 soc	NEW VOLUME [FAT	4096	2006-Nov-17 2:16:1(	2006-Nov-17 2:16:2(		no
14	2016	NEW VOLUME [FAT	4096	2017-Jan-01 12:01:3	2020-Jul-30 18:46:1(		no
15	vimal	NEW VOLUME [FAT	4096	2011-Jan-11 23:02:1	2011-Jan-11 23:02:2		no
16	dhs nov14 (0)	NEW VOLUME [FAT	4096	2020-Aug-07 19:34:3	2020-Aug-07 19:34:3		no
17	fACKUP~2	NEW VOLUME [FAT	0	2020-Aug-07 19:29:(	2020-Aug-07 19:29:(		yes
18	msg biodtat.pdf	NEW VOLUME [FAT	343297	2004-Jan-01 0:17:59	2004-Jan-01 0:09:36		no
19	DSC01479.JPG	NEW VOLUME [FAT	5195903	2004-Jan-01 0:05:56	2012-Sep-05 20:13:(		no
20	SREE BHAVANI stk	NEW VOLUME [FAT	378368	2014-Oct-03 11:15:2	2014-Oct-03 11:24:1		no
21	SREE BHAVANI stk	NEW VOLUME [FAT	378368	2014-Nov-01 9:08:12	2014-Nov-01 9:06:26		no
22	f.out	NEW VOLUME [FAT	1	2016-Jul-07 20:38:1(	2016-Jul-07 20:38:1(		yes

```
Created By AccessData® FTK® Imager 3.1.2.0

Case Information:
Acquired using: ADI3.1.2.0
Case Number: case007
Evidence Number: 007
Unique description: sdt
Examiner: Swarna
Notes:

---------------------------------------------------------------

Information for C:\Users\AI_LAB\Desktop\swarna case 1\Case007_evd007:

Physical Evidentiary Item (Source) Information:
[Device Info]
 Source Type: Physical
[Drive Geometry]
 Cylinders: 245
 Tracks per Cylinder: 255
 Sectors per Track: 63
 Bytes per Sector: 512
 Sector Count: 3,948,544
[Physical Drive Information]
 Drive Model: USB Device
 Drive Serial Number: 090326137c91b8
 Drive Interface Type: USB
 Removable drive: True
 Source data size: 1928 MB
 Sector count:    3948544
[Computed Hashes]
 MD5 checksum:    17bbd2f0c89656ae2168e8f40231e38a
 SHA1 checksum:   fe1019ab313c3feb5e36f85888e55b7d6f0a5ae2

Image Information:
 Acquisition started:   Thu Aug 06 13:43:31 2026
 Acquisition finished:  Thu Aug 06 13:45:35 2026
 Segment list:
  C:\Users\AI_LAB\Desktop\swarna case 1\Case007_evd007.E01
  C:\Users\AI_LAB\Desktop\swarna case 1\Case007_evd007.E02

Image Verification Results:
 Verification started:  Thu Aug 06 13:45:35 2026
 Verification finished: Thu Aug 06 13:45:40 2026
 MD5 checksum:    17bbd2f0c89656ae2168e8f40231e38a : verified
 SHA1 checksum:   fe1019ab313c3feb5e36f85888e55b7d6f0a5ae2 : verified
```

RESULT:

A forensic image of the storage device was successfully acquired and verified using FTK Imager. The generated hash values confirmed that the image maintained data integrity.

EXP 3 **INTRODUCTION TO DIGITAL FORENSICS (TRYHACKME)**

AIM:

To understand the basic concepts of digital forensics and learn how digital evidence is identified, collected, preserved, and analyzed using the TryHackMe learning platform.

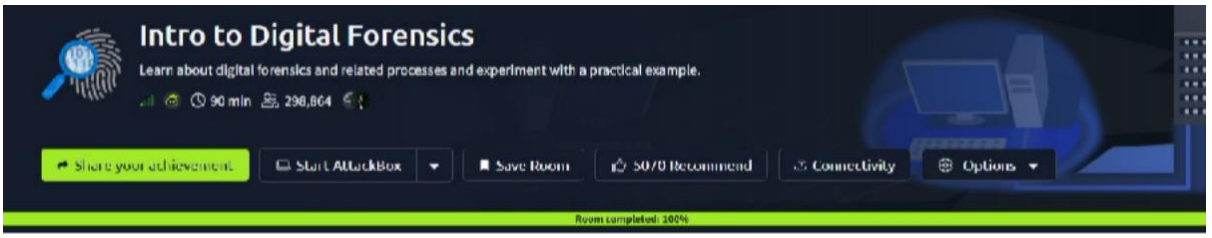
TOOLS REQUIRED:

1. TryHackMe platform
2. Web browser
3. Internet connection

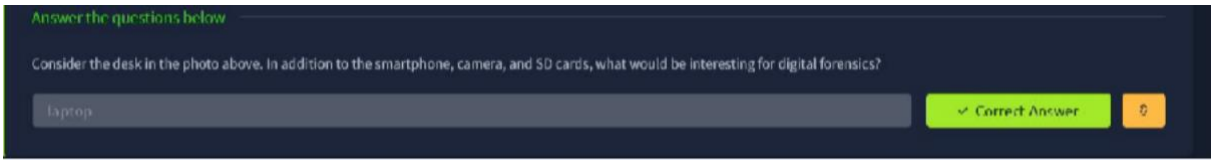
PROCEDURE:

1. Open a web browser and visit the TryHackMe website.
2. Sign in to your TryHackMe account.
3. Search for the **Introduction to Digital Forensics** room.
4. Open the room and read the introduction to digital forensics.
5. Study the importance of digital evidence in an investigation.
6. Learn how digital evidence is collected and preserved without altering the original data.
7. Understand the role of forensic investigators in examining digital devices and files.
8. Read the room's learning materials and complete the tasks provided.
9. Answer the questions based on the information given in the room.
10. Review the completed tasks and note the important concepts learned.

OBSERVATION:









Answer the questions below

It is essential to keep track of who is handling it at any point in time to ensure that evidence is admissible in the court of law. What is the name of the documentation that would help establish that?

Chain of Custody

✓ Correct Answer

Task 4 Practical Example of Digital Forensics

Everything we do on our digital devices, from smartphones to computers, leaves traces. Let's see how we can use this in the subsequent investigation.

Download Task Files

Our cat, Gado, has been kidnapped. The kidnapper has sent us a document with their requests in MS Word Document format. We have converted the document to PDF format and extracted the image from the MS Word file for your convenience.

You can download the attached file to your local machine for inspection; however, for your convenience we have added the files to the AttackBox. To follow along, open the terminal on the AttackBox, then go to the directory `/root/Rooms/introdigitalforensics` as shown below. In the following terminal output, we changed to the directory containing the case files.

Terminal

```
root# cd /root/Rooms
root# cd introdigitalforensics
root# ls
letter-image.jpg  ransom-letter.doc  ransom-letter.pdf  ransom-lettter-2.zip
```

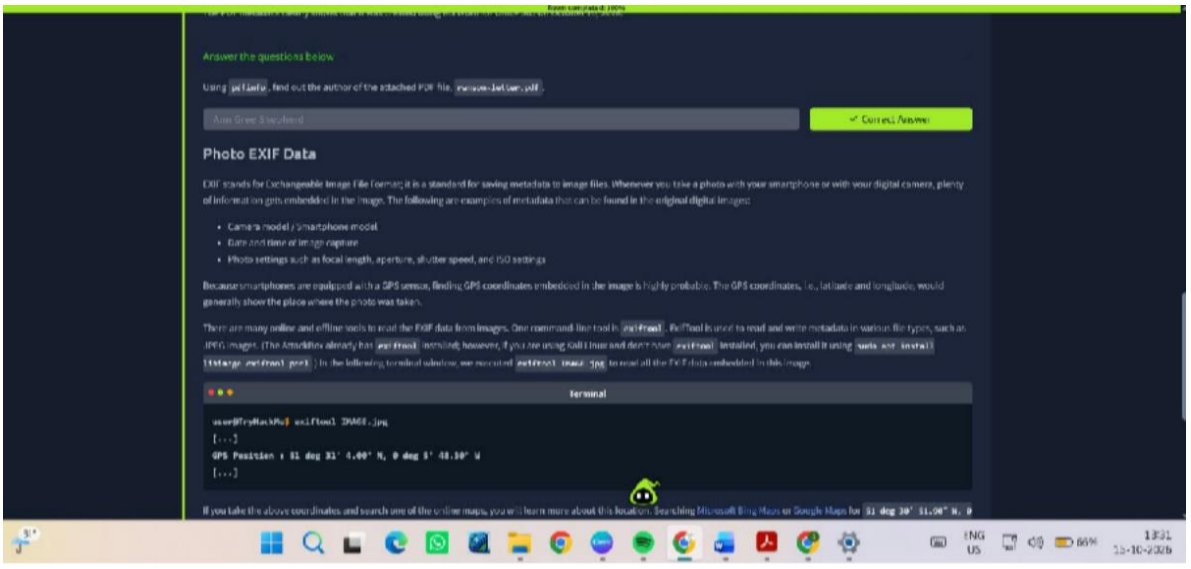
Document Metadata

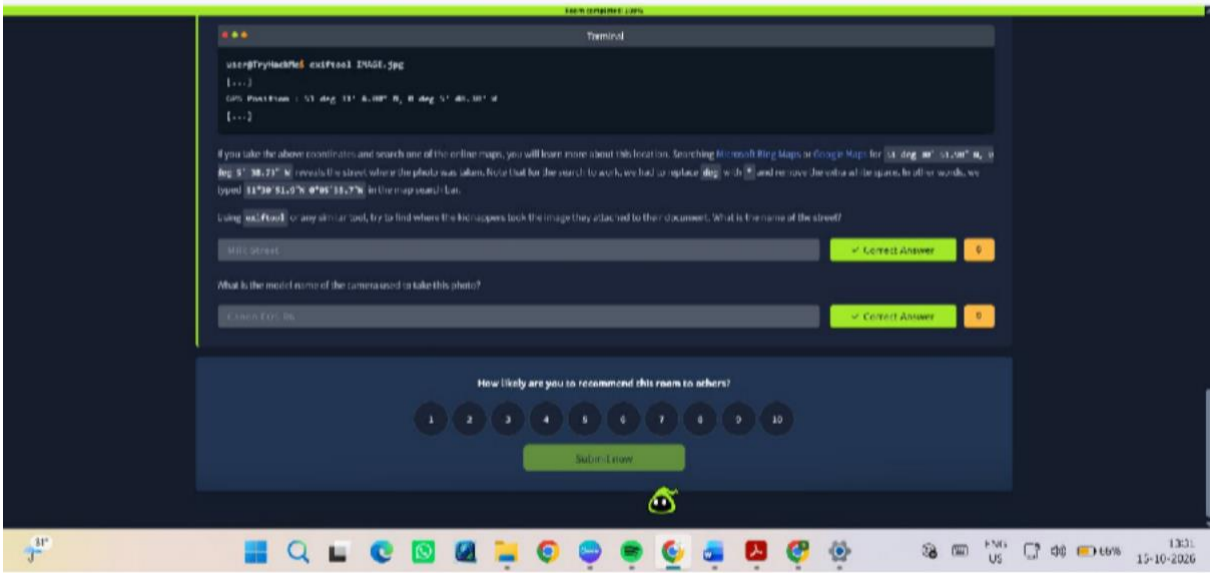
When you create a text file, `TXT`, some metadata gets saved by the Operating System, such as file creation date and last modification date. However, much information gets kept within the file's metadata when you use a more advanced editor, such as MS Word. There are various ways to read the file metadata; you might open them within their official viewer/editor or use a suitable forensic tool. Note that exporting the file to other formats, such as `PDF`, would maintain most of the metadata of the original document, depending on the PDF writer used.

Let's see what we can learn from the PDF file. We can try to read the metadata using the program `pdfinfo`. Pdfinfo displays various metadata related to a PDF file, such as title, subject, author, creator, and creation date. (The AttackBox already has `pdfinfo` installed; however, if you are using Kali Linux and don't have `pdfinfo` installed, you can install it using `sudo apt install poppler-utils`.) Consider the following example of using `pdfinfo DOCUMENT.pdf`.

Terminal

```
user@TryHackMe$ pdfinfo DOCUMENT.pdf
Creator:        Microsoft® Word for Office 365
```





The following concepts were studied:

- Meaning and importance of digital forensics.
- Identification and collection of digital evidence.
- Preservation of evidence and maintaining its integrity.
- Examination and analysis of digital evidence.
- Role of digital forensics in investigating cybercrimes.

RESULT:

The Introduction to Digital Forensics room on TryHackMe was studied to understand the basic concepts, procedures, and importance of digital forensics in cybercrime investigations.

EXP4:

Find Last Connected USB on Your System (USB Forensics)

AIM:
To identify the last connected USB device on the system using USBDeview.

PROCEDURE:

1. Open the browser, go to the NirSoft website and download USBDeview (zip file).
2. Extract the downloaded ZIP file to any folder.
3. Right click on USBDeview.exe and select "Run as administrator".
4. The USBDeview main window lists all USB devices that are currently connected and previously used.
5. Click on the "Last Plug/Unplug Date" column header to sort the entries. The top entry is the most recently used USB.
6. Note down the Device Name, Description, Serial Number and Last Plug/Unplug Date.
7. Double click on the device to view detailed information.
8. Export the report from the File menu (optional).
9. Record the Device Name, Description, Serial Number, Last Plug/Unplug Date, Vendor ID and Product ID in the lab manual.

OBSERVATION:

```
------------------------------------------------------
Device Name          : Dell KB216 Wired Keyboard
Description          : USB Composite Device
Device Type          : Unknown
Connected            : Yes
Safe To Unplug       : Yes
Disabled             : No
USB Hub              : No
Drive Letter         :
Serial Number        :
Registry Time 1      : 13-08-2026 13:23:39
Registry Time 2      : 21-02-2025 10:39:20
VendorID             : 413c
ProductID            : 2113
Firmware Revision    : 33.09
WCID                 :
USB Class            : 00
USB SubClass         : 00
USB Protocol         : 00
Hub / Port           :
Computer Name        : DESKTOP-EP0A4JR
Vendor Name          :
Product Name         :
ParentId Prefix      : 6&2456173 0&0
Service Name         : usbccgp
Service Description: @usb.inf,%GenericParent.SvcDesc%;Microsoft USB Generic Parent Driver
Driver Filename      : usbccgp.sys
Device Class         :
Device Mfg           : (Standard USB Host Controller)
Friendly Name        :
Power                : 100 mA
USB Version          : 2.00
Driver Description: USB Composite Device
Driver Version       : 10.0.26100.8972
Driver InfSection    : Composite.Dev.NT
Driver InfPath       : usb.inf
Instance ID          : USB\VID 413C&PID 2113\5&2fcc3441&0&8
Capabilities         : Removable, SurpriseRemovalOK
Install Time         : 21 02 2025 10:39:20
First Install Time: 21-02-2025 10:39:20
Connect Time         : 13-08-2026 13:23:39
Disconnect Time      :
------------------------------------------------------
```

RESULT:

The last connected USB device on the system was identified using USBDeview.

EXP 5 How to View Last Activity of Your PC

AIM:

To view the last activity of the PC using LastActivityView.

PROCEDURE:

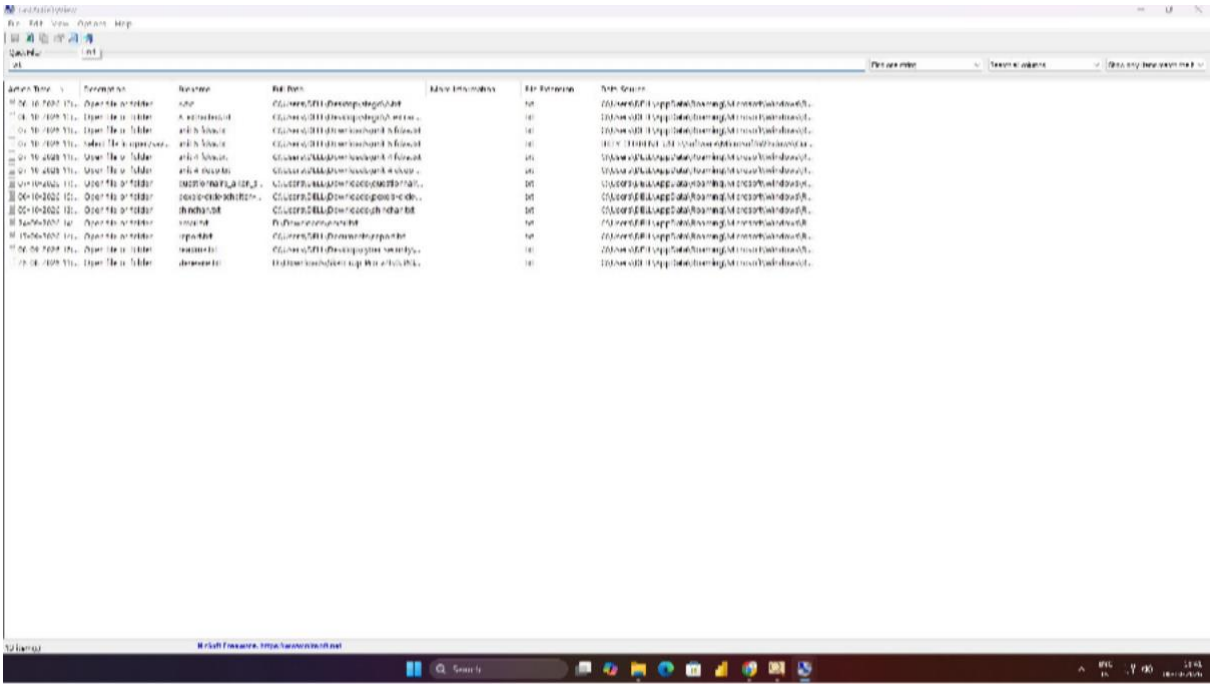
1. Open the browser, go to the NirSoft website, search for "LastActivityView" and download the ZIP file.
2. Extract the ZIP file using WinRAR/7-Zip to any folder.
3. Double click on "LastActivityView.exe" (no installation is required).
4. Observe the main window, which shows the Action Time, Description and Filename / Full Path of the recent activities.
5. Use the Filter option from the Options menu to view specific types of activities (optional).
6. Double click on any activity entry to view its detailed information.
7. Save the report from the File menu in HTML / CSV / Text / XML format.
8. Note down the last executed program, last opened file, system start time, installed software and other activity.
9. Analyze the collected information and close the tool.

OBSERVATION:

==========

Action Time : 20-08-2026 13:31:02
Description : Run .EXE file
Filename : AUDIODG.EXE
Full Path : C:\WINDOWS\SYSTEM32\AUDIODG.EXE
More Information : Microsoft Corporation, Microsoft® Windows® Operating System, Windows Audio Device Graph Isolation, 10.0.26100.1 (WinBuild.160101.0800)
File Extension : EXE
Data Source : C:\WINDOWS\Prefetch\AUDIODG.EXE-AB22E9A6.pf

==========



RESULT:

The last activities of the PC were viewed and recorded using LastActivityView.

EXP 6 How to Extracting Browser Artifacts

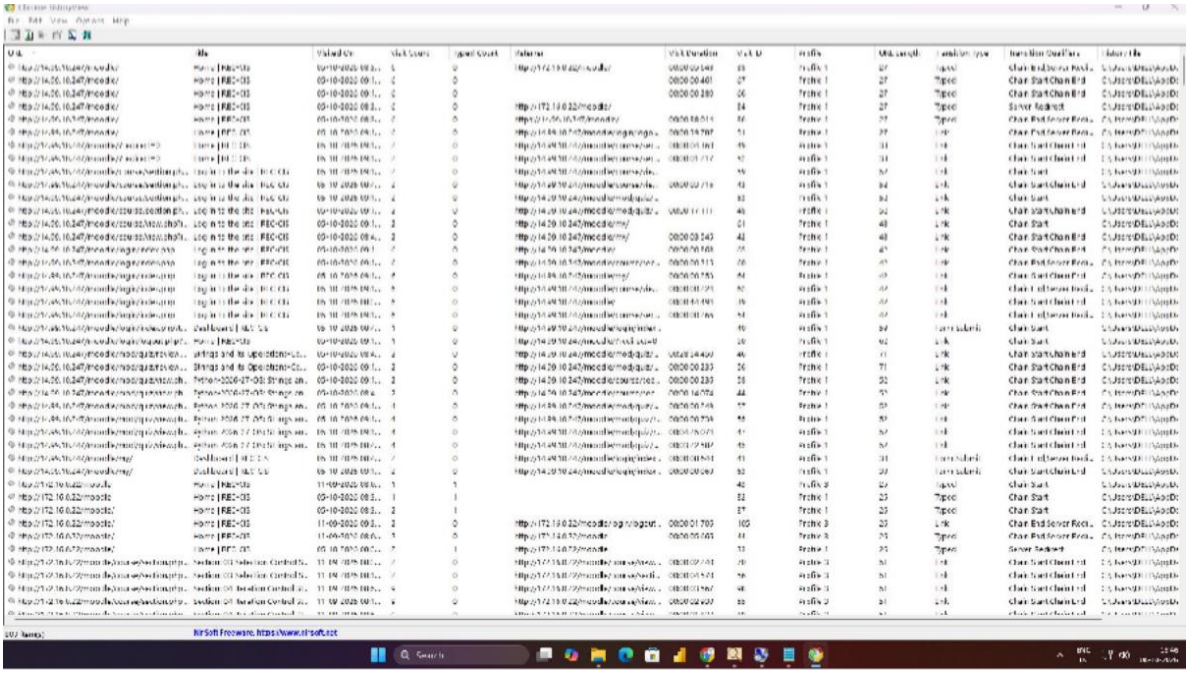
AIM:

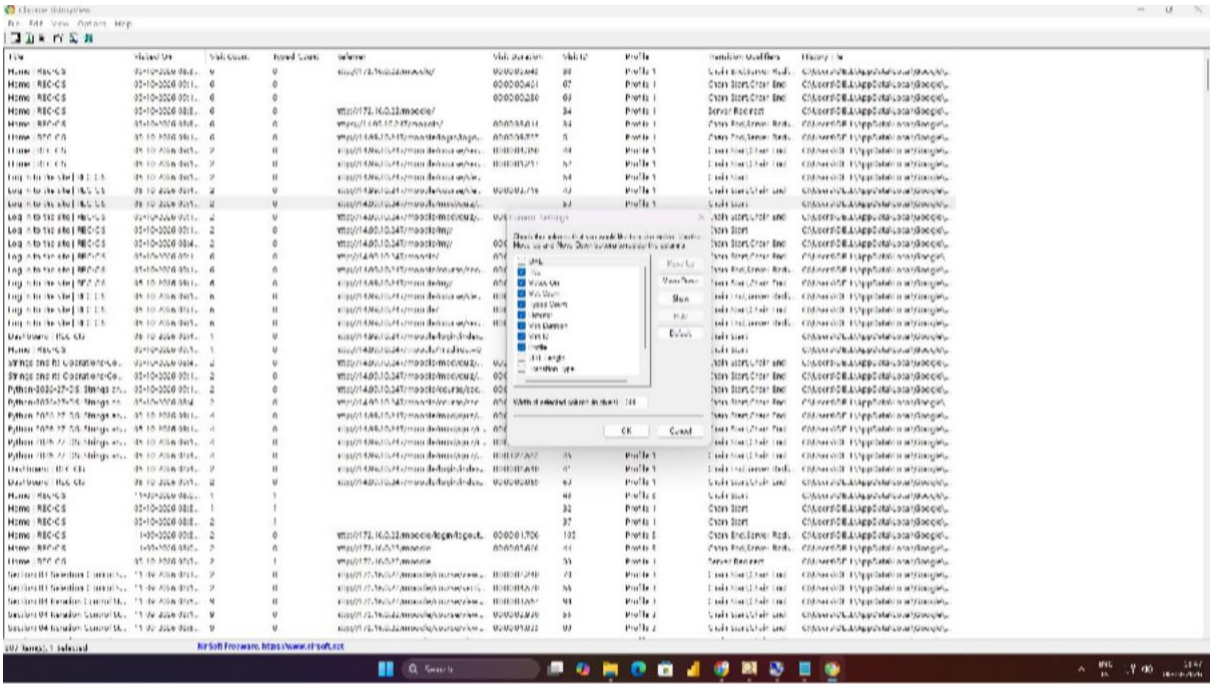
To extract browser artifacts using ChromeHistoryView.

PROCEDURE:

1. Open the browser, go to the NirSoft website, search for "ChromeHistoryView" and download it.
2. Extract the ZIP file using WinRAR/7-Zip to any folder.
3. Double click on "ChromeHistoryView.exe" (no installation is required). The tool automatically loads the Chrome history.
4. Observe the main window, which lists all the visited web pages.
5. Click the View menu and select the columns to display (URL, Title, Visit Time, Visit Count, Typed Count, Referrer, Visit ID).
6. Export the data from the File menu in CSV / HTML / Text format (optional).
7. Analyze the visited websites and identify the frequently visited sites.
8. Record the most visited site, total number of visits, suspicious URL (if any), date and time, and referrer website in the observation sheet.
9. Close the tool.

OBSERVATION:





RESULT:

The browser artifacts were extracted and analyzed using ChromeHistoryView.

How to View Last Activity of Your PC

AIM:

To view the last activity of the PC using LastActivityView.

PROCEDURE:

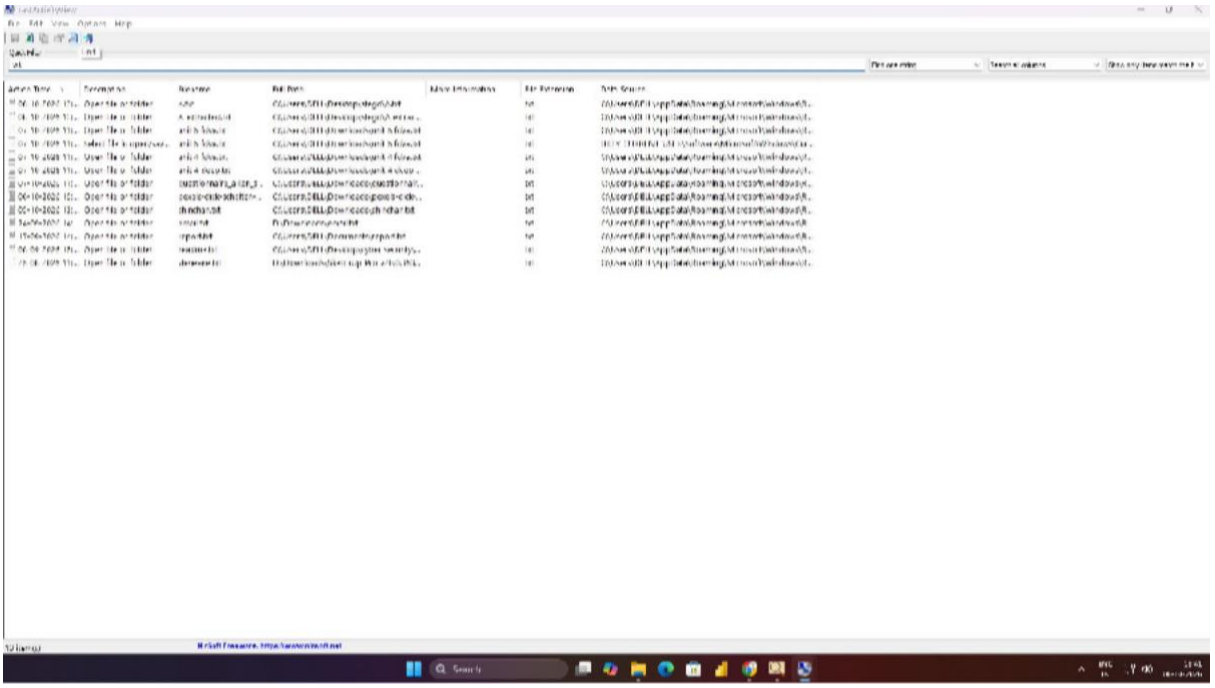
1. Open the browser, go to the NirSoft website, search for "LastActivityView" and download the ZIP file.
2. Extract the ZIP file using WinRAR/7-Zip to any folder.
3. Double click on "LastActivityView.exe" (no installation is required).
4. Observe the main window, which shows the Action Time, Description and Filename / Full Path of the recent activities.
5. Use the Filter option from the Options menu to view specific types of activities (optional).
6. Double click on any activity entry to view its detailed information.
7. Save the report from the File menu in HTML / CSV / Text / XML format.
8. Note down the last executed program, last opened file, system start time, installed software and other activity.
9. Analyze the collected information and close the tool.

OBSERVATION:

==========

Action Time : 20-08-2026 13:31:02
Description : Run .EXE file
Filename : AUDIODG.EXE
Full Path : C:\WINDOWS\SYSTEM32\AUDIODG.EXE
More Information : Microsoft Corporation, Microsoft® Windows® Operating System, Windows Audio Device Graph Isolation, 10.0.26100.1 (WinBuild.160101.0800)
File Extension : EXE
Data Source : C:\WINDOWS\Prefetch\AUDIODG.EXE-AB22E9A6.pf

==========



RESULT:

The last activities of the PC were viewed and recorded using LastActivityView.

EXP 8 How to Collect Email Evidence in Victim PC

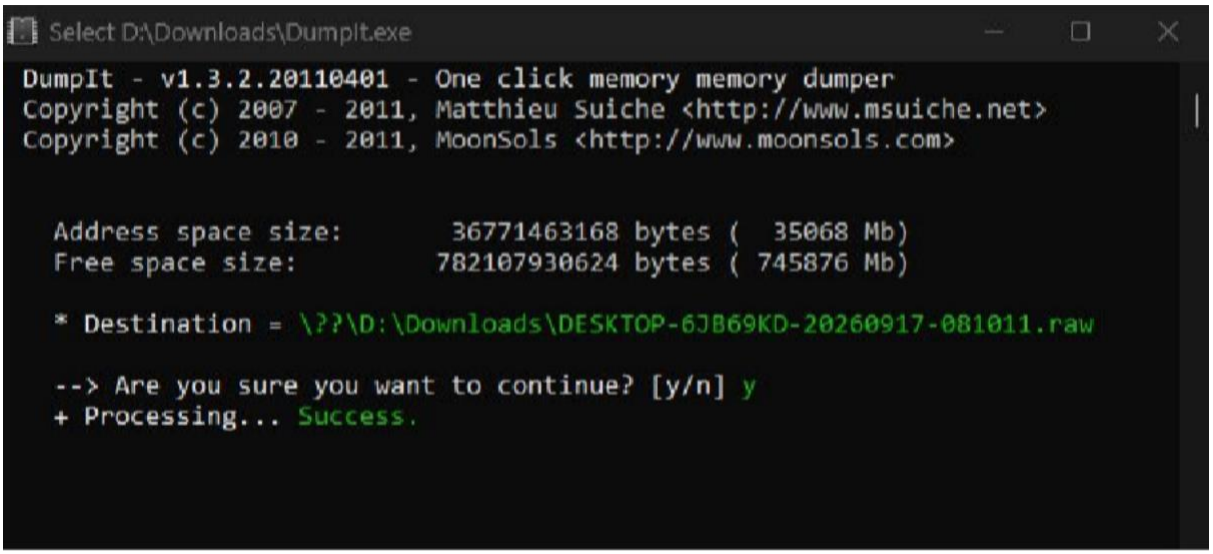
AIM:

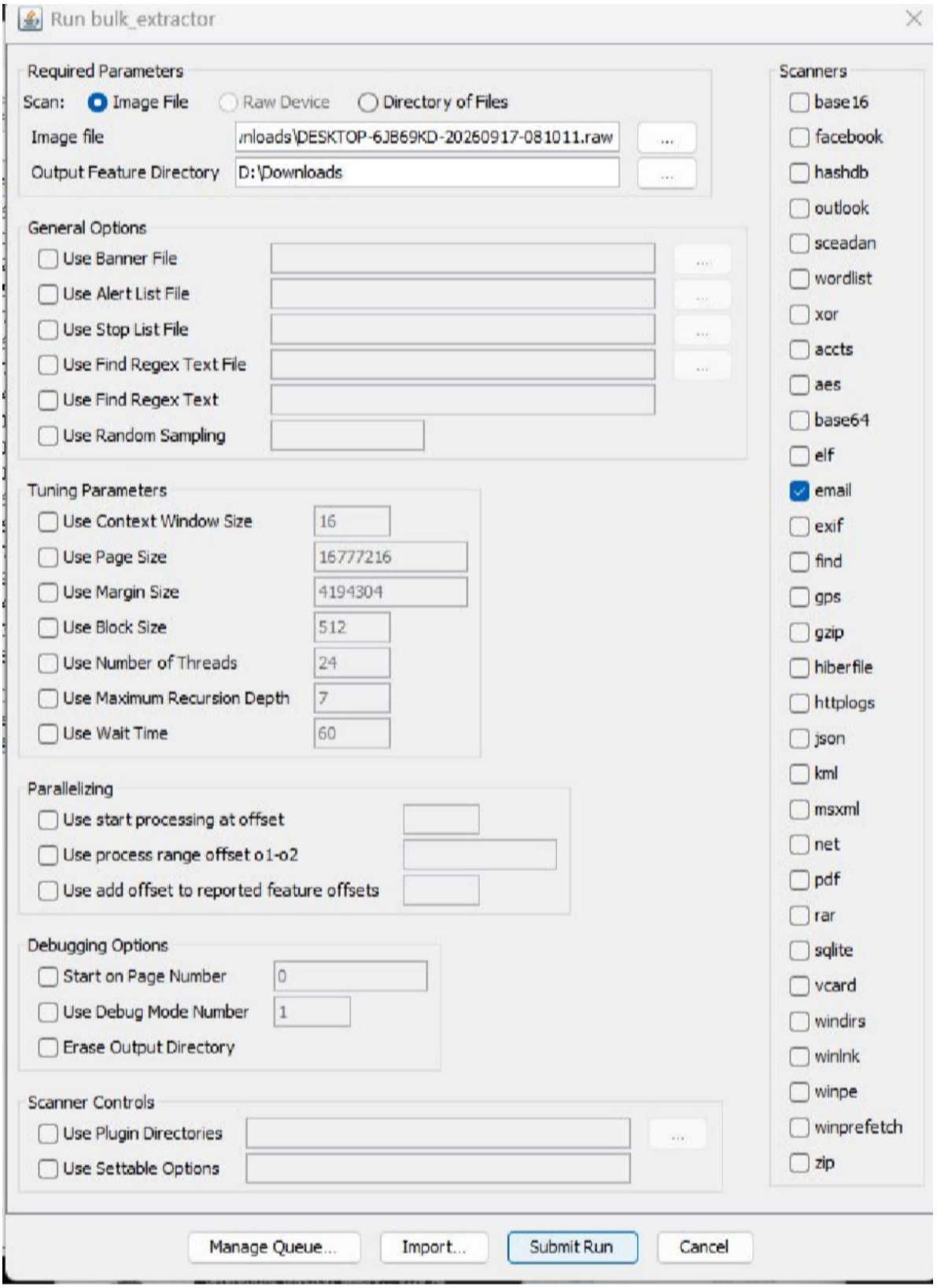
To collect email evidence in a victim PC using DumpIt.

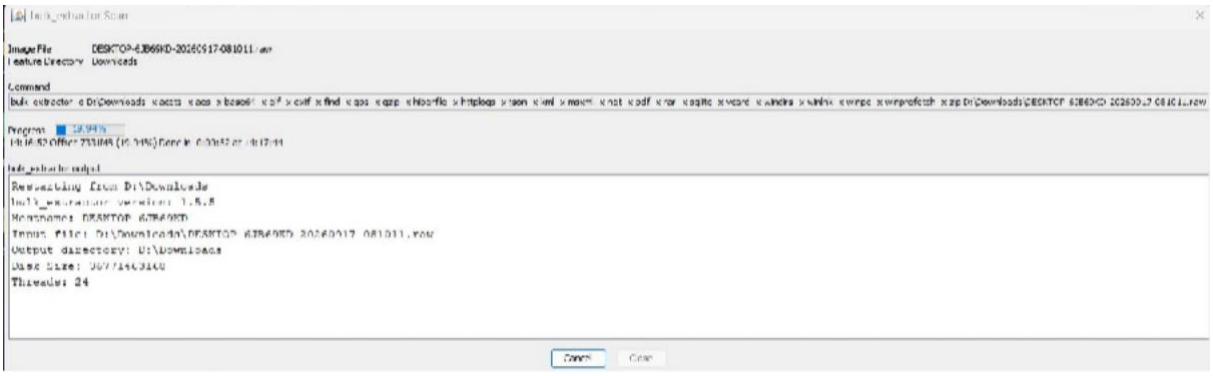
PROCEDURE:

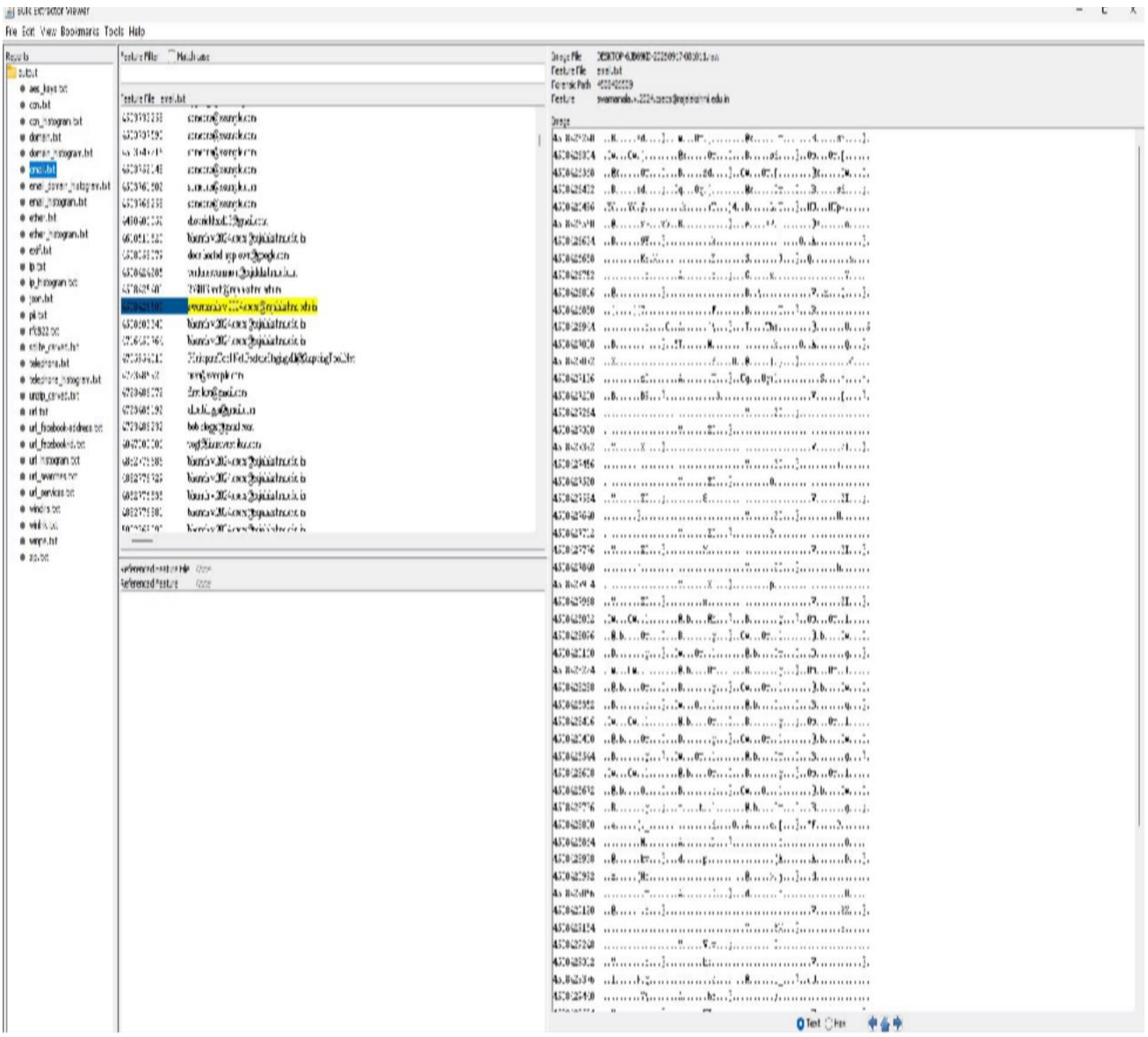
1. Open the browser, go to the MoonSols DumpIt page and download DumpIt.
2. Extract the downloaded file to get DumpIt.exe.
3. Right click on DumpIt.exe and select "Run as administrator".
4. Click "Yes" in the confirmation message to start capturing the RAM.
5. Wait until the memory dump is completed and the success message is displayed.
6. Locate the memory dump file (memory.raw) in the same folder as DumpIt.exe.
7. Copy the memory.raw file to a secure external drive without modifying or renaming it.
8. Analyze the memory dump using memory forensics tools like Volatility / Rekall, for example: volatility -f memory.raw --profile=Win10x64 pslist
9. Note down the date and time of collection, system information, DumpIt version, file name and size, and collected by in the evidence log sheet.
10. Close the DumpIt window.

OBSERVATION:




Run bulk_extractor
Required Parameters
Scan: Image File Raw Device Directory of Files
Image file
/nloads\DESKTOP-6JB69KD-20260917-081011.raw
Output Feature Directory
D:\Downloads
General Options
Use Banner File
Use Alert List File
Use Stop List File
Use Find Regex Text File
Use Find Regex Text
Use Random Sampling
Tuning Parameters
Use Context Window Size 16
Use Page Size 16777216
Use Margin Size 4194304
Use Block Size 512
Use Number of Threads 24
Use Maximum Recursion Depth 7
Use Wait Time 60
Parallelizing
Use start processing at offset
Use process range offset o1-o2
Use add offset to reported feature offsets
Debugging Options
Start on Page Number 0
Use Debug Mode Number 1
Erase Output Directory
Scanner Controls
Use Plugin Directories
Use Settable Options
Scanners
base16
facebook
hashdb
outlook
sceadan
wordlist
xor
accts
aes
base64
elf
email
exif
find
gps
gzip
hiberfile
httplogs
json
kml
msxml
net
pdf
rar
sqlite
vcard
windirs
winlnk
winpe
winprefetch
zip
Manage Queue...
Import...
Submit Run
Cancel





RESULT:

The physical memory of the victim PC was captured using DumpIt and the memory dump file was collected for email evidence analysis.

EXP 9 Comparison of Two Files for Forensics Investigation Using Compare IT Software

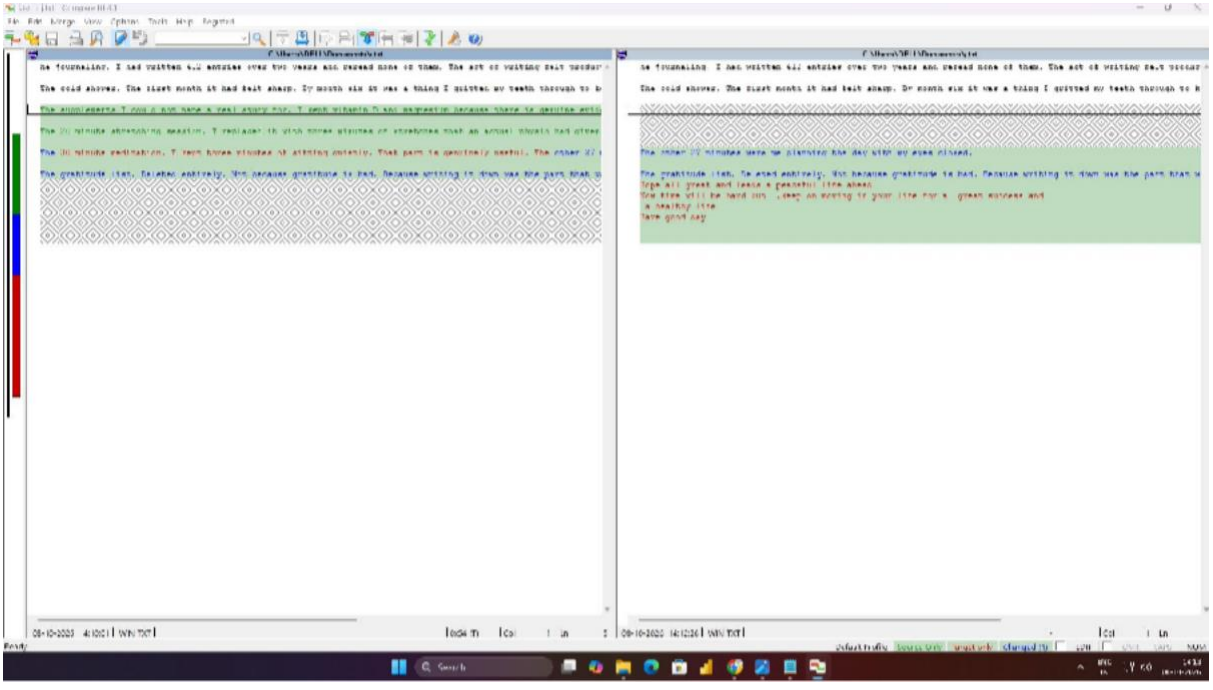
AIM:

To compare two files for forensics investigation using Compare IT software.

PROCEDURE:

1. Download Compare It from the official source and install it on the system.
2. Launch Compare It using the icon on the desktop.
3. Click the left side "Folder" icon, select the first file (File_A.txt) and click Open.
4. Click the right side "Folder" icon, select the second file (File_B.txt) and click Open.
5. Click Compare to start the comparison.
6. View the two files side by side with the colored differences (Green – Added, Red – Deleted, Yellow – Modified).
7. Analyze the highlighted differences.
8. Edit the files directly in the comparison window if needed (optional).
9. Click the Print / Report icon to generate the differences report (optional).
10. Close the tool.

OBSERVATIONS:



RESULT:

Two files were compared and the differences between them were identified using Compare IT.

EXP:10

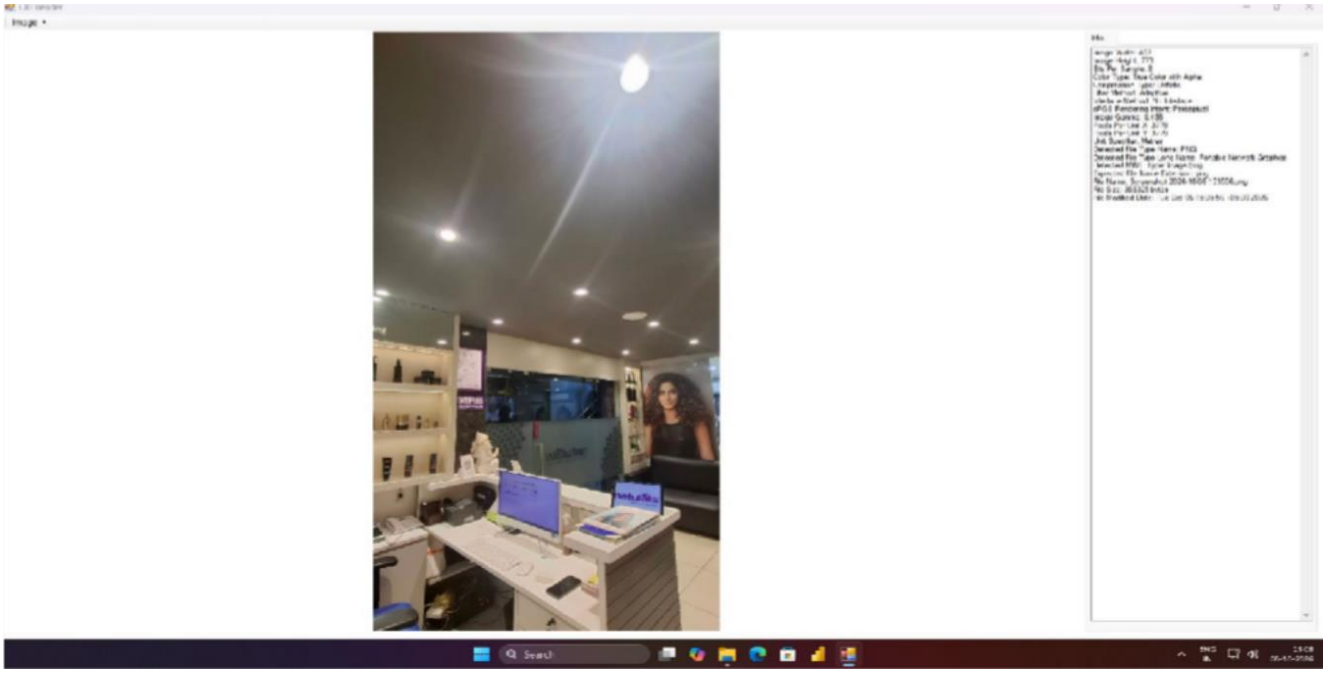
How to Extract Exchangeable Image File Format (EXIF) Data from Image Files using ExifReader Software

AIM:
To extract Exchangeable Image File Format (EXIF) data from image files using ExifReader software.

PROCEDURE:

1. Open the browser, go to the official ExifReader website and download the latest version.
2. Run the downloaded setup file (ExifReader.exe), follow the installation wizard, click Next and then Install.
3. Click Finish to complete the installation.
4. Double click on the ExifReader icon to launch it.
5. Click the File menu, select "Open Image File...", select the image file (for example sample.jpg) and click Open.
6. View the EXIF data displayed for the image (Make, Model, Date/Time Original, GPS Latitude, GPS Longitude, Software, etc.).
7. Click File → "Save Report..." and save the report as a .txt or .html file (optional).
8. Analyze the date/time, GPS location, camera make and model, exposure settings and software.
9. Note down the important EXIF details in the EXIF findings sheet.
10. Close the application.

Observation



RESULT:

The EXIF data was extracted from the image file and analyzed using ExifReader.